

CS360 Nano Quiz - 25.02.26(Wed)

1. Please choose a **WRONG** explanation about DBMS (1.5 minute).

- (1) Database is a collection of data that is managed by a DBMS.
- (2) DBMS is one of the most complex software.
- (3) Efficiency is very important for DBMS.
- (4) DBMS is widely used in business area, but not in scientific investigations.
- (5) DBMS should enable durability, i.e., the recovery of database from failures.

2. Please choose a **WRONG** explanation about data models (2 minute).

- (1) Relational model always achieves high performance.
- (2) Document-oriented model has high flexibility, but low functionality and low complexity.
- (3) Key-value model can achieve high performance, high scalability, and high flexibility.
- (4) Column-oriented model usually is better than relation model in terms of performance, scalability, and flexibility, but worse than relational model in terms of complexity and functionality.
- (5) Graph model has a higher or at least similar complexity than the above data models.

3. Please choose a **WRONG** explanation about data systems (2 minute).

- (1) There is no single perfect database system.
- (2) Major components of modern data systems include database systems, caches, search indexes, stream processing, and batch processing.
- (3) Data systems always MUST combine all components of database systems, caches, search indexes, stream processing, and batch processing.
- (4) If the database is changed due to the write request, the application code should invalidate or update cache in order to keep the cache in sync with the primary database, and also, apply the updates to search index as well.

4. Please choose a **WRONG** explanation about DBMS (1.5 minute).

- (1) There are two kinds of commands: DML and DDL.
- (2) There are two kinds of DML statements: query processing and transaction processing.
- (3) Query compiler only parses an input SQL query to a parse tree, not generate a query plan.
- (4) Query processing requires not only query compiler component, but also other components such as execution engine, buffer manager, and storage manager.

CS360 Nano Quiz - 25.03.05(Wed)

1. Please choose a **WRONG** explanation about data model and relational data model (2 minutes).

- (1) Data model is described in terms of three parts: structure of data, operations on data, constraints on data.
- (2) Operations on data for relational model is relational algebra.
- (3) In relational model, a database is a set of relations, and a relation is a set of tuples.
- (4) Tuples are also called as objects, records, or rows.
- (5) The number of attributes is called as cardinality.

2. Please choose a **WRONG** explanation about schema and instance (1.5 minutes).

- (1) Schema of a relation is a combination of the name of the relation and the set of attributes of the relation.
- (2) Database schema is the set of the schemas of the relations in a database.
- (3) A relation changes over time, and the current set of tuples of a relation is called as an instance.
- (4) The schema of a relation is static and not changed over time.
- (5) Even if we reorder the set of tuples of a relation, the resulting relation is equivalent to the original relation.

3. Please choose a **WRONG** explanation about a key of a relation (2 minutes).

- (1) For any two tuples A and B in a relation, the key attribute values of A and B are different with each other.
- (2) A key of a relation should be a single attribute.
- (3) If a key is CHAR(10), and the key value of a tuple is "DATABASE", then the last two characters of the key value is padded.
- (4) A key of a relation is declared when the relation is created.
- (5) A value of a key might be NULL, if the key is declared as UNIQUE.

4. Please choose a **CORRECT** explanation about relational algebra (2 minutes).

- (1) The power of SQL is equivalent to the power of general programming languages.
- (2) Relational algebra always returns a relation as output.
- (3) Less powerfulness of SQL is good for DBMS, but not good for users.
- (4) Only SQL is used in DBMS. The relational algebra is not used in DBMS at all.

CS360 Nano Quiz - 25.03.10(Mon)

1. Please choose a **WRONG** explanation about relational algebra operations (2 minutes).

- (1) Relations R and S must have the same schema in order to apply set operations on them.
- (2) Projection of R returns a relation that has a subset of R's columns.
- (3) Selection of R returns a relation that has a subset of R's tuples.
- (4) We denote the cardinality of R as $|R|$. Then, $|R \times S| = |R| + |S|$, where 'x' is Cartesian product.
- (5) We denote the number of attributes of R as $[R]$. Then, $[R \times S] = [R] + [S]$, where 'x' is Cartesian product.

2. Please choose a **WRONG** explanation about relational algebra operations (2 minutes).

- (1) Natural join may generate dangling tuples.
- (2) We denote the cardinality of R as $|R|$. Then, $|R \text{ nj } S| \leq |R| * |S|$, where 'nj' is a natural join.
- (3) Theta-join with a condition C produces the same result with Cartesian product and selection with the condition C.
- (4) An expression tree is evaluated in a top-down manner.
- (5) An expression tree can be represented in a linear notation

3. Please choose a **WRONG** explanation about relational algebra operations (2 minutes).

- (1) After applying the renaming operation to R, R's instance does not change.
- (2) There is usually more than one relational algebra expression for the same query.
- (3) Six basic operations are union, intersection, selection, projection, product, and renaming.
- (4) Intersection, theta-join and natural join can be expressed by using six basic operations.
- (5) An expression tree can be expressed multiple assignment statements using temporary relations.

4. Please choose a **WRONG** explanation about referential integrity constraints (2 minutes).

- (1) In our example DB, all 'starName' column values of StarsIn are a subset of 'name' column values of MovieStar.
- (2) Similarly, all 'movieYear' column values of StarsIn are a subset of 'year' column values of Movies.
- (3) Similarly, all 'producerC#' column values of Movies are a subset of 'cert#' column values of MovieExec.
- (4) Some 'title' column values of Movies may not be present in 'movieTitle' column values of StarsIn.
- (5) Referential integrity constraints and key constraints refer to the same constraints.

CS360 Nano Quiz - 25.03.12(Wed)

1. Please choose a **WRONG** explanation about functional dependencies (1.5 minutes).

- (1) It is a generalization of the idea of a key for a relation.
- (2) It is denoted as $A_1, A_2, \dots, A_n \rightarrow B_1, B_2, \dots, B_m$.
- (3) For the above notation, we say that $A_1, A_2, \dots, A_n$ functionally determine $B_1, B_2, \dots, B_m$.
- (4) A's attributes and B's attributes must appear consecutively in a relation schema.

2. Please choose a **WRONG** explanation about a key (2 minutes).

- (1) A key of a relation R functionally determines all other attributes of R.
- (2) A subset of a key cannot functionally determine all other attributes of R.
- (3) A superset of a key functionally determines all other attributes of R.
- (4) A relation has more than one primary key.
- (5) A relation may have more than one candidate key.

3. Please choose a **WRONG** explanation about reasoning about FD (2 minutes).

- (1) For FD's $A \rightarrow B$ and $B \rightarrow C$, adding a new FD $A \rightarrow C$ is a transitive rule.
- (2) Two sets of FD's S and T are equivalent if and only if S follows from T, and T follows from S.
- (3) For right sides, both splitting and combining rules are possible.
- (4) For left sides, only splitting rule is possible.

CS360 Nano Quiz - 25.03.17(Mon)

1. Please choose a **WRONG** explanation about functional dependencies (2 minutes).

- (1) Given a set of FD's S , any set of FD's equivalent to S becomes a basis for S .
- (2) Given a set of FD's S , there is only a single minimal basis.
- (3) If any FD is removed from a minimal basis, the result is no longer a basis.
- (4) The result of projecting functional dependencies is a minimal basis.

(2), (4) both answers are correct

2. Please choose a **WRONG** explanation about normal forms (2 minutes).

- (1) A prime attribute is a member of some candidate key of R .
- (2) R violates 3NF, if nonprime attribute of R is transitively dependent on the primary key of R .
- (3) The Movies1 relation in our example violates BCNF.
- (4) 3NF is more strict than BCNF.

CS360 Nano Quiz - 25.03.19(Wed)

1. Please choose a **WRONG** explanation about three properties of decomposition (2 minutes).

- (1) Elimination of anomalies means eliminating redundancy, update anomalies, and deletion anomalies.
- (2) Recoverability of information means that we can recover the original relation from the tuples in its decomposition.
- (3) Preservation of dependencies means that when we reconstruct the original relation from the decomposition by joining, the result satisfies the original FD's.
- (4) BCNF guarantees all the above three properties.
- (5) A decomposition would be better when it guarantees more properties.

2. Please choose a **WRONG** explanation about BCNF and 3NF (1.5 minutes).

- (1) BCNF can eliminate more redundancy than 3NF.
- (2) Both BCNF and 3NF give us "recoverability of information" commonly.
- (3) 3NF can give us "preservation of dependencies", because it is less strict than BCNF.
- (4) Lossless join is not possible, if we decompose a relation using 3NF.

3. Please choose a **WRONG** explanation about MVD and 4NF (2 minutes).

- (1) Relations in BCNF may still have a kind of redundancy.
- (2) If a relation is already in BCNF, but has MVD, then, the key of the relation is a set of all the attributes.
- (3) If a relation has three attributes, A,B, and C, and MVD $A \twoheadrightarrow B$ exists, then MVD $A \twoheadrightarrow C$ also exists.
- (4) The condition of 4NF is essentially the BCNF condition, but applied to MVD's instead of FD's.
- (5) 4NF can eliminate redundancy caused by MVD's, but cannot eliminate redundancy due to FD's.

CS360 Nano Quiz - 25.03.31(Mon)

1. Please choose a **CORRECT** explanation about converting relational algebra into Datalog query.

- (1) Selection operation is converted into only a single rule, regardless of the selection condition.
- (2) Theta join operation is converted into only two rules, regardless of the join condition.
- (3) A complex expression tree in relational algebra cannot be expressed in a Datalog query.
- (4) Datalog can express recursive queries that relational algebra cannot express.
- (5) Relational algebra can express queries that Datalog cannot express.

CS360 Nano Quiz - 25.04.02(Wed)

1. Please choose a **WRONG** explanation about physical pointer and logical pointer.

- (1) A physical pointer contains a value that refers a physical address in memory.
- (2) A logical pointer contains a FK value that refers a PK value of another relation in database space
- (3) Referential integrity can be achieved, when there are some dangling references.
- (4) An example of dangling references in web database is a broken link in HTML.
- (5) When specifying a FK in SQL DDL, you should use "FOREIGN KEY" and "REFERENCES".

2. Please choose a **WRONG** explanation about referential integrity options.

- (1) Default option is NO ACTION.
- (2) NO ACTION means that deletion or update of a tuple referred by other tuples is rejected by DBMS's.
- (3) CASCADE means deletion of all tuples that refer to deleted tuple.
- (4) SET NULL or SET DEFAULT means setting FK values of referencing tuples to NULL or default value.
- (5) CASCADE deletion is performed only within a single relation.

3. Please choose a **WRONG** explanation about three-valued logic.

- (1) The result of an arithmetic operator is NULL when we operate on a NULL value and any value.
- (2) The result of a comparison operator is UNKNOWN when we operate on a NULL value and any value.
- (3) We cannot use NULL explicitly as an operand in SQL.
- (4) x OR y is FALSE when x is UNKNOWN and y is FALSE.
- (5) x AND y is UNKNOWN when x is TRUE and y is UNKNOWN.

CS360 Nano Quiz - 25.04.07(Mon)

1. Please choose a **WRONG** explanation about join in SQL.

- (1) It is possible to ask a query that involves two or more tuples from the same relation.
- (2) We can list a relation R as many times as we need in FROM clause by using tuple variables.
- (3) A tuple variable is also called an alias.
- (4) If a join condition involves attributes with the same name in multiple relations, we need to disambiguate the attribute names by using <relation name>.<attribute name>.
- (5) Only a join between PK and FK can be used in WHERE clause.

2. Please choose a **WRONG** explanation about subqueries in SQL.

- (1) A subquery is surrounded by parentheses.
- (2) A subquery can also have another subquery.
- (3) Subqueries can be used in union, intersection, and difference.
- (4) A subquery can return either a single constant or a relation.
- (5) It is NEVER possible to transform a query containing subqueries into an equivalent flat query.

3. Please choose a **WRONG** explanation about SQL operators for a tuple s and a relation R.

- (1) The condition "EXIST R" is true, if and only if, R is not empty.
- (2) The condition "s IN R" is true, if and only if, s is equal to one of the values in R.
- (3) The condition "s NOT IN R" is true, if and only if, s does not match any value in R.
- (4) The condition "s > ANY R" is true, if and only if, s is greater than every value in R.
- (5) The condition "s > ALL R" is true, if and only if, s is greater than every value in R.

4. Please choose a **WRONG** explanation about correlated subqueries.

- (1) An uncorrelated subquery can be evaluated only once.
- (2) A correlated subquery is required to be evaluated many times.
- (3) We can use a subquery instead of a stored relation in FROM clause, without an alias.
- (4) If a subquery is correlated, then, whenever a tuple variable's value outside the subquery is assigned to some term in the subquery, the subquery must be evaluated.
- (5) When writing a correlated query, it is important to be aware of the scoping rules for names.

CS360 Nano Quiz - 25.04.09(Wed)

1. Please choose a **WRONG** explanation about joins between two relations R and S in SQL.

- (1) Cartesian product in relational algebra is expressed as "R CROSS JOIN S" in SQL.
- (2) Theta-join in relational algebra is expressed as "R JOIN S ON " in SQL.
- (3) Natural join in relational algebra is expressed as "R NATURAL JOIN S" in SQL.
- (4) Full outer 'natural' join in relational algebra is expressed as "R FULL OUTER JOIN S" in SQL.
- (5) Left outer theta join in relational algebra is expressed as "R LEFT OUTER JOIN S ON <condition>" in SQL.

2. Please choose a **WRONG** explanation about aggregation and group-by in SQL.

- (1) "GROUP BY" is followed by a list of grouping attributes.
- (2) Only attributes in GROUP BY can be used in SELECT, except aggregated attributes.
- (3) NULL value is ignored in any aggregation.
- (4) NULL is treated as an ordinary value when forming groups.
- (5) Result of every aggregation over an empty bag is NULL.
- (6) "HAVING <condition>" filters out groups that do not satisfy <condition>.

3. Please choose a **WRONG** explanation about DML.

- (1) Database Manipulation Language (DML) changes the state of DB.
- (2) The INSERT statement can insert only a single tuple at once.
- (3) In the DELETE statement, we cannot directly specify a tuple to be deleted. We have to specify a condition using the primary key in the WHERE clause.
- (4) A single UPDATE statement can update multiple attributes at once.
- (5) The UPDATE statement can update multiple tuples at once.